Indirect reciprocity with inertial reputation updating

Storyline 3, from information to judgement (final draft, for a multidisciplinary journal)

Introduction

From village communities to online marketplaces, people decide whom to help on the basis of what they know about others' past conduct[1,2]. Those who help gain a good reputation, and a good reputation attracts help from third parties. This feedback, known as indirect reciprocity, can sustain cooperation among strangers who may never meet again[3–6]. Theory has identified social norms under which it is evolutionarily stable[7–10], but these results assume that everyone agrees about who is good. In reality, each person forms a private view from noisy and incomplete observations[11,12], and once views diverge, most of these norms no longer sustain cooperation[13–16].

The trouble lies in punishment. Stable norms allow people to refuse help to those in bad standing without losing their own good standing, because the refusal is seen as justified[8,9,17]. Whether a refusal looks justified, however, depends on the observer's own view of the person refused. An observer who still regards that person as good condemns the punisher, and disagreement about one individual spreads to those who act towards it[13,18,19]. Those who punish defectors thus put their own reputation at risk, a punishment dilemma that undermines the sanctions on which cooperation depends[17,20,21]. The harm done by disputed judgements has two components, how often they occur and how much each one changes a reputation. The first depends mainly on what observers know, the second on how they revise what they believe.

Most proposed remedies act on what observers know. Public institutions, gossip and the synchronization of opinions bring observers to a common view[22–25], and the aggregation of several observations gives each observer more evidence to judge from[20]. Better information reduces disputed judgements, but it cannot remove the errors that make every judgement fallible[13,26]. What remains is a problem of judgement, namely how a standing verdict about another person should be revised as new evidence arrives. Any revision rule must balance two errors, condemning cooperators on disputed evidence and excusing defectors whose misconduct is not yet conclusive. Human institutions often strike this balance through procedure. They escalate sanctions only with repeated violations[27,28] and restore an offender's standing only after a period of good conduct[29], and people likewise revise their impressions of others asymmetrically[30–33]. Models of indirect reciprocity, by contrast, mostly revise reputations through a single symmetric rule, whether the latest assessment, a score or a fixed window of observations[9,13,20,34]. How such revision rules shape the punishment dilemma, and how they should be designed, remains open.

Here, we show that the way private judgements are revised can sustain cooperation despite the punishment dilemma, without any shared information. We develop a model of indirect reciprocity in which judgements have inertia (Fig. 1). An established reputation changes only after a run of contrary assessments, and the run required to lose standing can differ from the run required to regain it. While evidence accumulates, an individual under suspicion or on probation can already face partial consequences, depending on how strongly these intermediate states are expressed. Observers share no information and update on one observation at a time, so any agreement among them must emerge from how they revise their judgements.

We find that the direction of revision, rather than the amount of evidence it requires, decides whether private observers agree with one another and still tell cooperators from defectors. The beneficial direction mirrors the kind of misjudgement that private assessment produces under each norm, and rules selected solely by these reputational criteria make cooperation stable and easy to reach under four classical norms. The same amount of inertia placed in the wrong direction can eliminate cooperation altogether. Rules that protect cooperators from disputed evidence, however, can also shelter opportunists who defect only occasionally, a weakness shared by the aggregation and score-based rules we test. Letting suspicion carry partial consequences restores deterrence while cooperation remains high. Sustaining cooperation through private reputations is therefore not only a matter of what individuals know about each other, but also of how they revise what they believe.

Discussion additions

Two paragraphs to add to the Discussion. The first sets out the two routes out of the punishment dilemma. The second carries the core of Storyline 2.

The distinction between what observers know and how they revise what they believe offers a general way to organize remedies for the punishment dilemma. The harm that disputed judgements do to justified punishers depends on how often such judgements occur and on how strongly each one changes a reputation. Institutions, gossip, synchronization and generous assessment act on the first factor[22–24,35], whereas scores, aggregation and inertia act on the second[20,34]. The same separation appears in our theory. Near a resident population, the reputational cost of withholding help is the product of the assessment contrast, which measures how differently observers judge cooperation and defection, and the slope of the reputation map, which measures how strongly a change in assessments moves a reputation (Eq. (3)). The two routes are therefore complementary. Weak agreement can be offset by robust revision, as in our model, and fragile revision does little harm when information is public. Leniency, however, has a price on either route. Generous assessment and tolerant revision both protect deliberate defectors as well as justified punishers[35], as our comparison of inertia, aggregation and scores shows (Fig. 5a). A revision rule should therefore separate whether a single negative assessment can overturn a reputation, which inertia suppresses, from what it costs at the margin, which partial consequences restore. Disputed evidence should be costly but not catastrophic.

Our results also bear on how agreement about reputations arises. Agreement is usually attributed to mechanisms that align information across observers, such as institutions, gossip and opinion synchronization[22–24], or to observers who draw on several observations at once[20]. In our model, observers share nothing and update on one observation at a time, yet they largely agree. Two independent observers who each regard an individual as good with probability r disagree with probability $2r(1-r)$, so agreement follows whenever private judgements become reliable. Under the punishment dilemma, reliability is not automatic, because the assessments an individual receives depend on what observers think of its recipients. Inertia produces it when it is placed in the direction that separates the assessments of cooperators from those of defectors. Otherwise observers can agree on the wrong verdict, as under one-step Shunning, where they regard nearly everyone as bad (Fig. 2). Agreement through private revision complements information sharing rather than replacing it. It remains available where shared information is costly or untrusted, as in polarized societies[20,36], and it can be combined with institutions or gossip where they exist.

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